MBAI 5110: Big Data Systems Design

A Lambda Architecture Implementation for Predicting Bike Theft: A Case Study on Toronto Bicycle Theft

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Project Summary

This project showcases the design and implementation of a Lambda Architecture-based big data pipeline aimed at analyzing and predicting bicycle theft risk in Toronto. The goal of this project is to utilize the Bicycle Thefts dataset from the City of Toronto Open Data Portal (1975-2025) for the implementation of a Lambda Architecture-based big data pipeline focused on analyzing and predicting bicycle theft risk in Toronto. By integrating batch processing for historical analysis and model training with real-time streaming for live prediction, the system aims to provide actionable insights to reduce the bike thefts in Toronto. A Random Forest classifier is utilized to categorize theft risk levels (Low, Medium, High), with careful attention given to preventing data leakage. This solution illustrates how scalable data systems can deliver valuable insights for both city stakeholders and individual users.

Dataset

The analysis utilizes the Bicycle Thefts dataset from the City of Toronto Open Data Portal (1975–2025).

- **Data Source**: City of Toronto Open Data Portal
- **Raw Data Size**: (39,849, 35)
- **Post-Data Wrangling Size**: (31,559, 33)

Data Pipeline Architecture

The architecture diagram below illustrates Lambda Architecture designed to analyze bike theft patterns and provide real-time risk assessments. This structure encompasses two phases: Phase 1 and Phase 2. Phase 1 is Batch Processing ("Cold") and Phase 2 is Real-time streaming ("Hot"). This structure effectively bridges the gap between historical "Cold" data and live "Hot" data

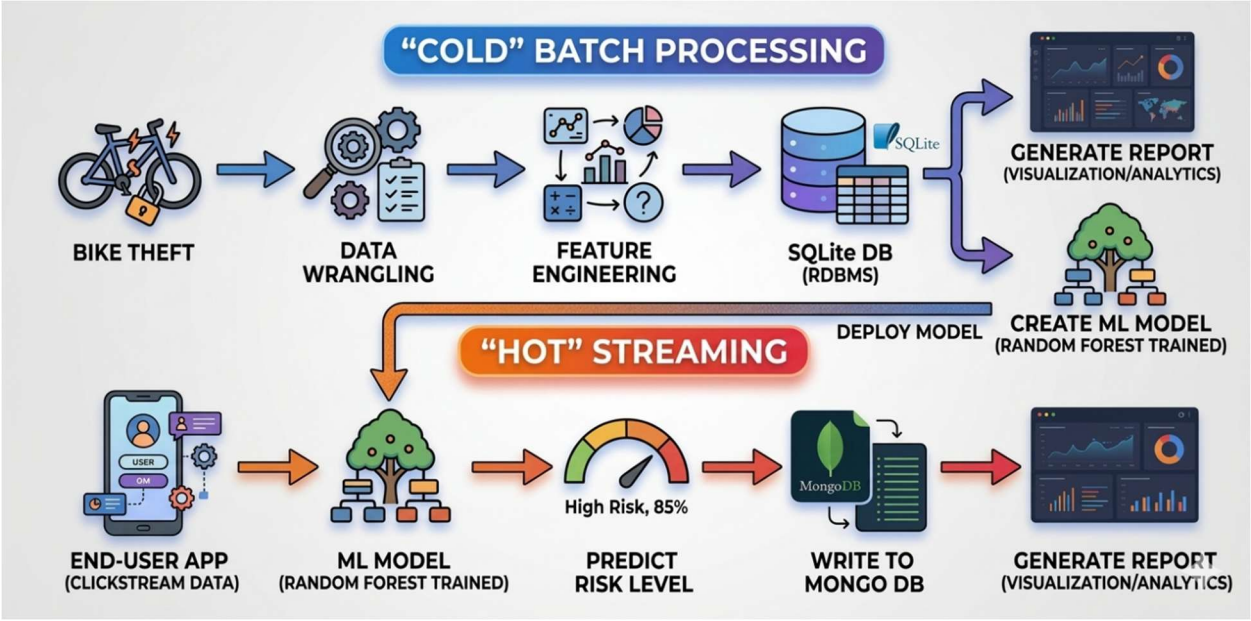

Figure 1- Big Data Pipeline

Phase 1: The "Cold" Path (Batch Processing)

This architectural layer comprises six stages designed for the high-capacity processing of historical datasets. In this context, precision and robustness are prioritized over real-time latency. The main objective is the refinement of the "Master Dataset" to facilitate high-accuracy predictive modeling and analytical reporting.

1. **Data Ingestion:** The initial phase involves the collection of the raw bike theft primary dataset.

2. **Data Wrangling and Pre-processing:** To ensure data integrity, the system performs comprehensive cleaning, including the identification and removal of duplicate records and the mitigation of missing values via statistical imputation or record exclusion. This process is done via the DataCleaning.ipynb notebook.

3. **Feature Engineering:** Raw variables are transformed into high value features- such as *Occurrence Period* and *Categorical Risk Levels*, to enhance the model's predictive power. The refined output is then persisted as a structured Cleaned_Bikes.csv file.

4. **Relational Database Management (RDBMS):** The processed dataset is ingested into SQLite to facilitate complex relational queries. This environment is optimized for structured analysis and the generation of standardized reports based on historical data.

5. **Exploratory Data Analysis (EDA):** A rigorous evaluation of "cold" data is conducted to identify underlying patterns:

    - **Problem Identification:** The "5 Ws" framework (Who, What, When, Where, Why) is applied to make the dataset interpretable and actionable.

    - **Visual Analytics:** Historical trends are converted into reports via the DataVisualization_RDMS.ipynb file.

6. **Predictive Modeling (Random Forest):** In this final phase, a Random Forest algorithm is trained to identify indicators of theft based on features derived (and other columns) during the exploratory analysis. To ensure the model's generalizability and prevent overfitting the master dataset is partitioned using a standard 80:20 train-test split ratio.

The Bridge: Random Forest Model

The Random Forest Model acts as the core of the system. Once trained in the Cold Path, it is deployed to the Hot Path for live decision-making.

Phase 2: The “Hot” Path (Real-Time Streaming)

This architectural layer comprises five distinct stages designed for the ingestion and real-time processing of high velocity and high-volume data originating from the end-user application. The pipeline is engineered for minimal latency to facilitate instantaneous risk level assessments.

1. **Ingestion (Clickstream Data):** The client-side application captures data points (including bike colour, cost, and manufacturer specifications) which are continuously streamed into the predictive pipeline.

2. **Inference Engine (Random Forest):** A pre-trained Random Forest model performs real-time classification on incoming records. By leveraging a high-dimensional feature set, the engine generates immediate predictive outputs.

3. **Real-Time Predictive Output:** During this stage, each record is subjected to instantaneous categorical assignment, resulting in a risk-level classification (e.g., “High Risk”).

4. **Persistent Storage:** To maintain high throughput without compromising application performance, predictive outcomes are persisted in MongoDB. This NoSQL database is specifically utilized for its ability to handle semi-structured, high-speed data streams.

5. **Visualization (Serving Layer):** In the final phase, raw predictive outputs are transformed into actionable intelligence. The system delivers interactive dashboards and analytical reports, providing stakeholders with a robust framework for monitoring theft trends in real time.

Feature Engineering

To bridge the gap between raw data ingestion and model inference, three targeted feature engineering strategies were implemented, focusing on Spatial, Temporal, and Categorical dimensions of the Toronto Bike Theft dataset. These transformations convert high-variance data into structured, high-signal features, ensuring the Random Forest model can effectively identify patterns.

Neighborhood Risk Categorization

The transformation of raw theft counts into a categorical Risk Profile for each neighborhood allows for intuitive analysis of geographic safety trends and provides structured input for predictive modeling.

Methodology: Quantile-Based Segmentation

Neighborhoods are categorized using data-driven thresholds (33rd and 66th percentiles) for more relative classification:

- **High Risk**: Top 33% of neighborhoods by theft counts.
- **Medium Risk**: Middle 33%.
- **Low Risk**: Bottom 33%.
- **Unknown**: Areas with no recorded thefts.

Key Technical Insights

1. **Normalization**: Reduces the impact of extreme outliers, enhancing the analysis's robustness.
2. **Handling Sparsity**: Identifying neighborhoods with zero recorded thefts as “Unknown” ensures data integrity for future reports.
3. **Scalability**: Utilizing approxQuantile provides efficient scaling across large datasets while maintaining accuracy.

Business Value

This feature enables stakeholders to quickly identify high-priority zones, allowing for targeted patrols based on risk categorization.

Temporal Period Classification

Categorizing 24-hour incident timestamps into four distinct Occurrence Periods simplifies daily crime cycle analysis and helps allocate security resources effectively.

Methodology: Hour-Based Binning

The raw OCC_HOUR data is mapped into four key categories according to common daily activity patterns:

- Morning (06:00 - 12:00)
- Afternoon (12:01 - 17:00)
- Evening (17:01 - 19:59)
- Night (20:00 - 05:59)

Key Technical Insights

1. **Logical Consistency**: Ensures all hours are accounted for, preventing data loss.
2. **Strategic Weighting**: Groups the lowest-light hours to identify peak theft windows.
3. **Efficiency**: High-level grouping facilitates easier visualization.

Business Value

This feature confirms peak theft times, enabling management to optimize security coverage according to hourly needs.

High-Cardinality Reduction (Bike Make/Model)

To manage the diverse range of bike manufacturers and models, high-cardinality distributions are simplified to focus on statistically significant entries.

Methodology: Top- N Filtering

Restricts entries to the 100 most frequent Makes and Models, categorizing any not on the list as “OTHER” to minimize noise.

Key Technical Insights

1. **Noise Reduction**: Concentrating on prevalent brands allows for focused analysis.

2. **Dimensionality Control**: Prevents unnecessary computational complexity during model training.

3. **Data Integrity**: Ensures clean schema for final analyses.

Business Value

This transformation allows for a clear "Market Share of Theft" analysis, highlighting key target brands without distractions.

Machine Learning

Prior to model training, all features representing post-incident data- specifically those relating to the timing of the theft (occ_, *report_*), police administrative boundaries (Division), and the legal classification of the crime (primary_offence)- were systematically removed to prevent target leakage. This ensured that the predictive analysis relied solely on precursor variables available to a user before an incident occurs. Subsequently, two machine learning models were developed using a Random Forest classifier to evaluate and predict bike theft risk levels.

Model A: With Location Features

This model was trained using all available location-based features.

Performance:

Accuracy: 100.00%

Precision: 100.00%

F1 Score: 100.00%

Confusion Matrix Summary (Model A)

The perfect classification indicates an over-reliance on spatial features, introducing potential data leakage

Note: The numerical labels represent the following risk categories: 0.0 (High Risk), 1.0 (Low Risk), and 2.0 (Medium Risk).

Label	Prediction	Count
0.0	0.0	410

1.0	1.0	353
2.0	2.0	394

Table 2- Confusion Matrix for Model A Risk Level Predictions

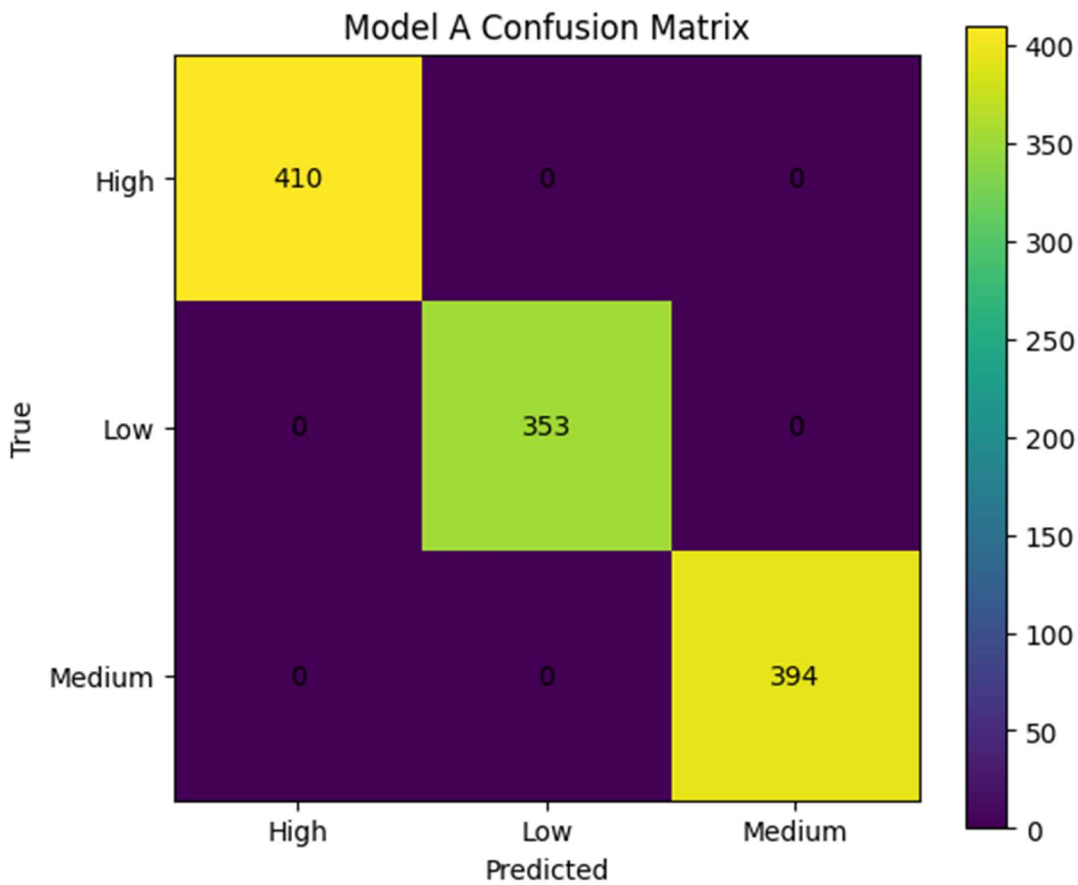

Figure 6- Heatmap of Confusion Matrix (Model A)

Model B: Without Location Features

By excluding location variables, Model B focuses on behavioral and temporal data.

Performance:

- Accuracy: 88.76%
- Precision: 90.48%
- F1 Score: 88.62%

Confusion Matrix Summary (Model B)

Model B provides a more realistic assessment of predictive performance. Without direct geographic indicators, the model must rely on underlying patterns in the data, such as temporal trends and other contextual features.

Note: The numerical labels represent the following risk categories: 0.0 (High Risk), 1.0 (Low Risk), and 2.0 (Medium Risk).

Label	Prediction	Count
0.0	1.0	3
0.0	0.0	299
0.0	2.0	108
1.0	1.0	350
1.0	2.0	3
2.0	0.0	6
2.0	2.0	378
2.0	1.0	10

Table 3- Confusion Matrix for Model B Risk Level Predictions

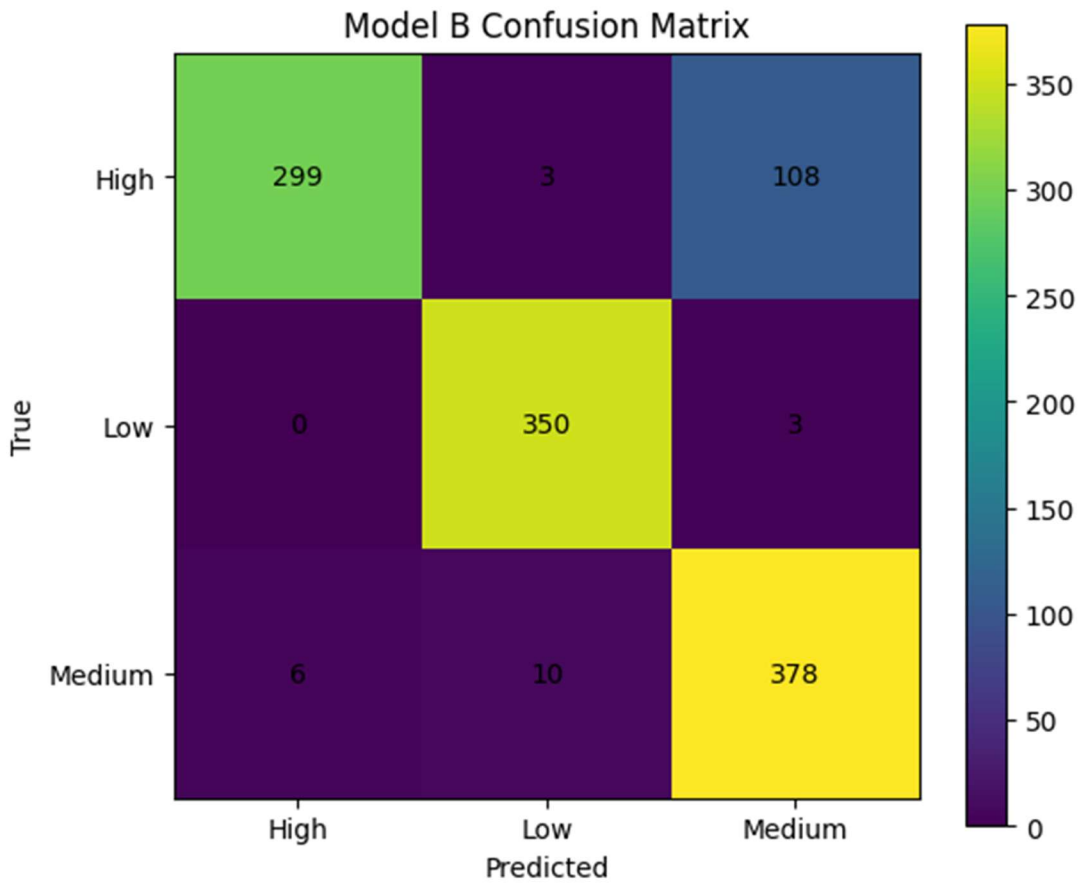

Figure 7- Heatmap of Confusion Matrix (Model B)

Model B Summary

While this model shows strong performance, it relies on underlying data patterns and avoids direct geographic indicators, making it suitable for real-world deployment.

Model Selection

Although Model A achieved perfect accuracy, it is unsuitable for deployment due to its data leakage. In contrast, Model B demonstrates robust performance and generalizability, making it the preferred choice for applications involving new data.

Simulating Real-Time Ecosystem: Clickstream Data Generation

To validate scalability and real-time processing capabilities, a synthetic data generator was created using PySpark, simulating thousands of concurrent users interacting with the application.

Key Simulation Characteristics

1. **Concurrent User Simulation**: 25,000 unique events simulate high-traffic scenarios.
2. **Contextual Realism**: Ensures statistical integrity by using actual values from the cleaned dataset.
3. **Temporal & Geospatial Mapping**: Events are categorized based on time-of-day logic and relevant police division data.
4. **Synthetic Reporting Lag**: Simulates realistic administrative delays for testing purposes.

Purpose in the Project

This stream serves as a stress test, measuring the risk-prediction model's performance under load and ensuring rapid risk assessments.

Key Insights

The key insights derived from the Historical Data Analysis reveal several noteworthy patterns:

1. **Temporal Insights**:
    - **Nighttime is Peak Time**: With 8,613 reported thefts, the night period slightly outnumbers morning and afternoon incidents. This indicates that thieves prefer the cover of darkness and reduced foot traffic for their activities. This is illustrated by *Chart 1: Bike Thefts by Time of Day* and *Chart 2: Bike Thefts by Hour of Day*

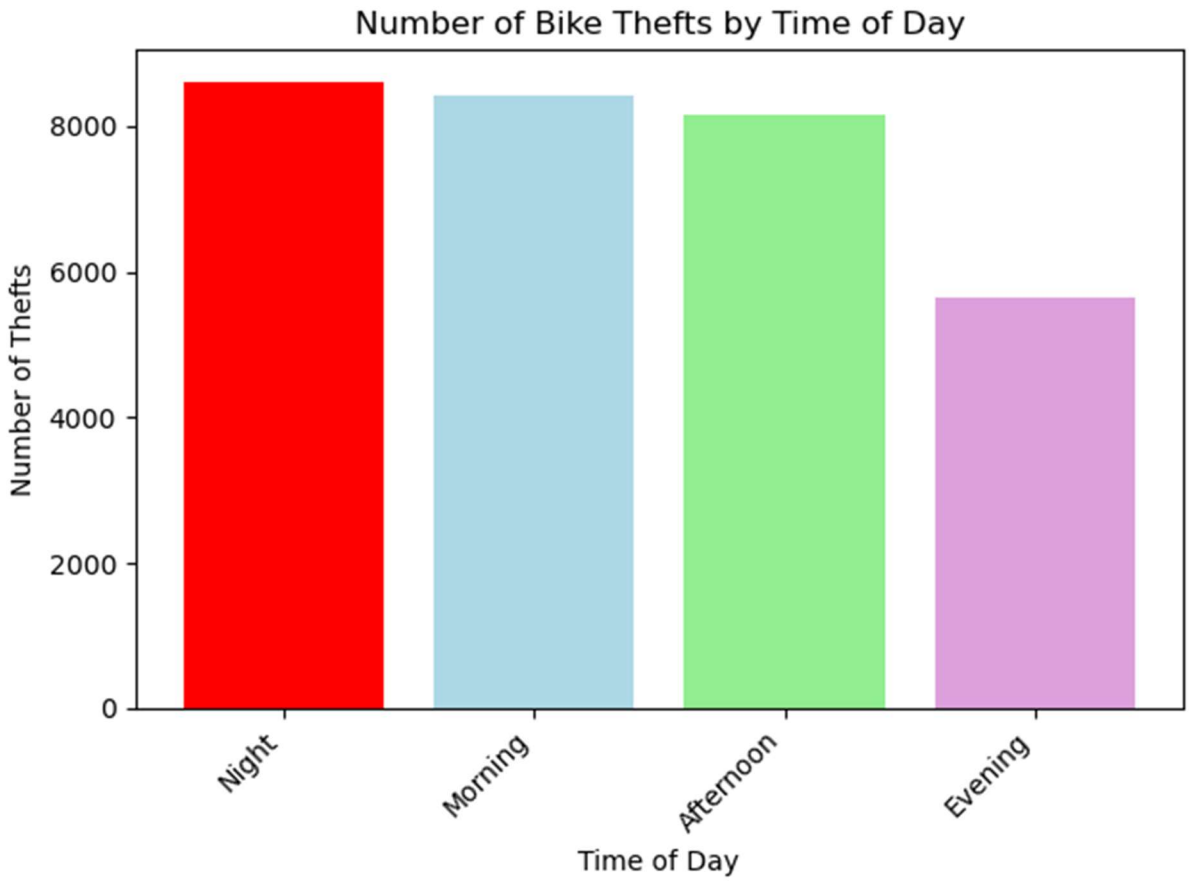

Chart 1- Bike Theft by Time of Day

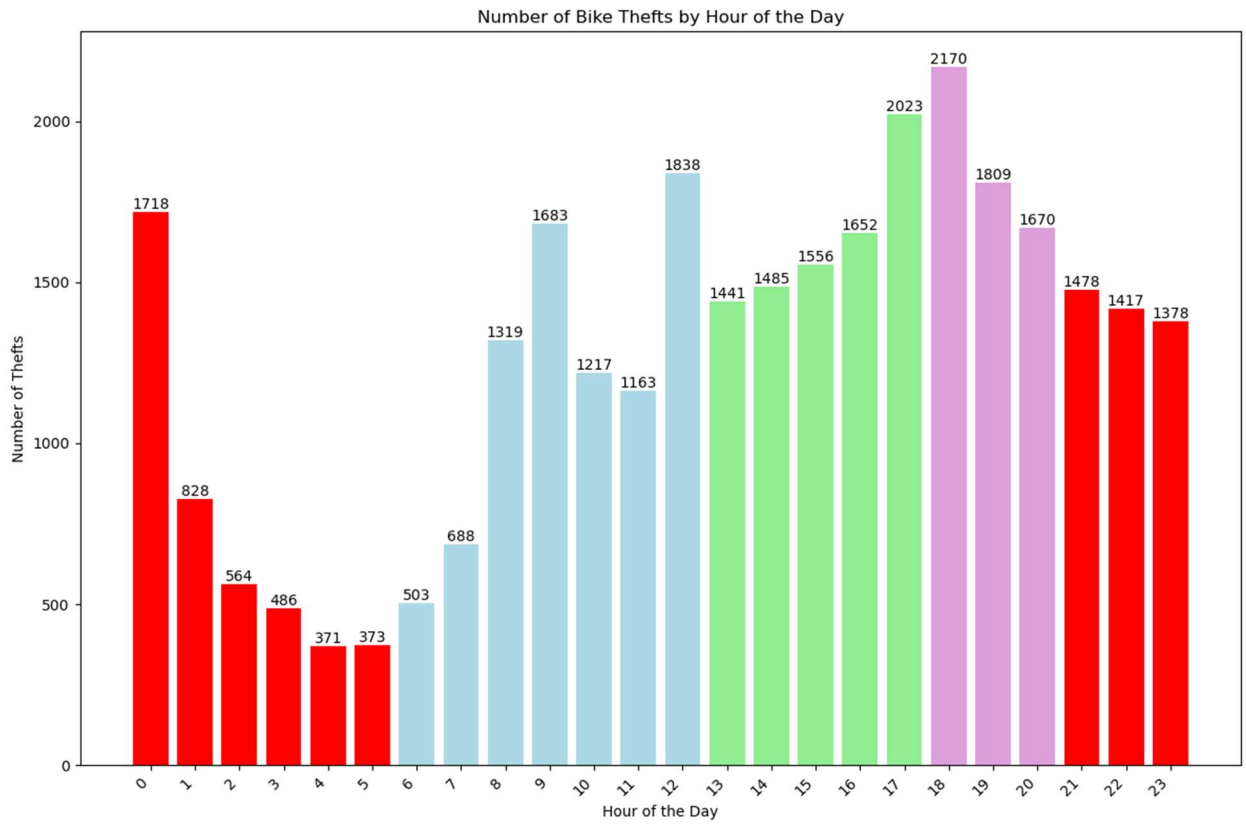

Chart 2- Bike Thefts by Hour of Day

2. **Spatial Insights**:
    - In the night category, thefts from Apartments/Condos (2,479) and Single Homes (1,865) are dominant for bicycle thefts, highlighting the risks associated with private structures like sheds and garages (995), which are

more frequently targeted than bars or restaurants (245). This is illustrated in *Chart 3*- Bike Thefts in the Night by Location Type

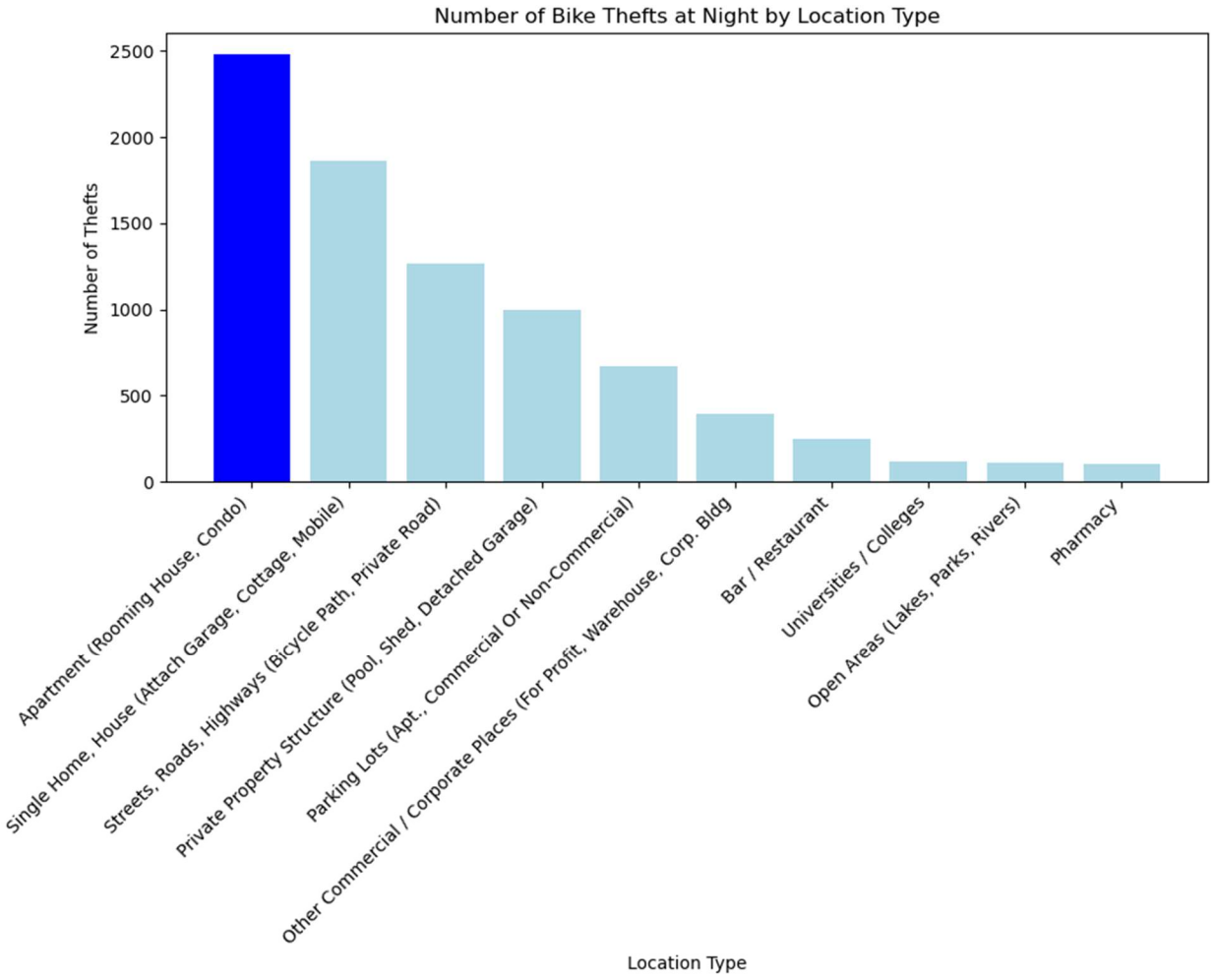

Chart 3- Bike Thefts in the Night by Location Type

3. **Geographic Hotspots**:
    - The Downtown Core shows a clear cluster of thefts, particularly in the Waterfront Communities (e.g., St. Lawrence, Harbourfront), likely due to dense modern condo developments with expansive underground parking.

This is illustrated by *Chart 4- Concentration of Bike Thefts at Night in Apartment Complexes by Neighbourhood*.

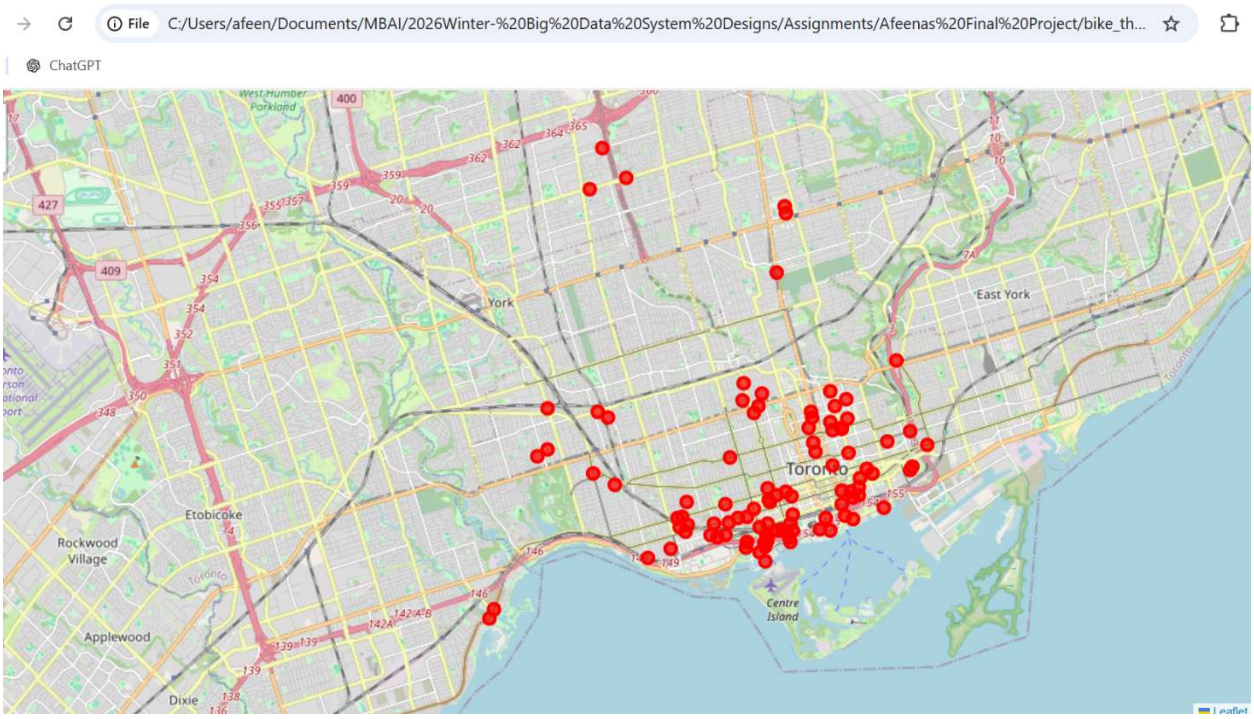

Chart 4- Concentration of Bike Thefts at Night in Apartment Complexes by Neighbourhood

Location	Bike Thefts
St Lawrence-East Bayfront-The Islands (166) Waterfront Communities-The Island (77)	147
Wellington Place (164) Waterfront Communities-The Island (77)	140
Harbourfront-CityPlace (165) Waterfront Communities-The Island (77)	135
Annex (95) Annex (95)	130
Fort York-Liberty Village (163) Niagara (82)	121

Table 3: Top 5 neighbourhoods by apartment-night theft count

Business Problem

The data indicates that despite residents investing in “secure” housing (condos and apartments), these locations account for the highest theft losses, leading to several secondary challenges:

- **Property Management Liability**: Heightened pressure on building managers to enhance surveillance and access controls.
- **Urban Mobility Barriers**: High theft rates in dense areas discourage residents from considering bicycles as viable transportation, contradicting city-wide “Green” initiatives.
- **Resource Misallocation**: Police efforts often focus on public areas, while data suggests that the frontline of bike theft has shifted to private spaces like parking garages.

Proposed Solutions and Strategic Implementation

Based on the data analysis and predictive modeling results, a multi mitigation strategy is proposed. This approach divides interventions into Municipal (City) Solutions and User-Centric (Individual) Solutions.

Municipal Solutions: Data-Driven Resource Allocation

Analytical findings indicate that theft incidents peak significantly during nocturnal hours, with a concentrated surge at 18:00 (2,170 recorded incidents). Furthermore, the spatial analysis identifies apartment complexes as high-volatility environments during the window between 18:00 and 00:00.

- **Strategic Recommendation:** It is proposed that municipal law enforcement and private security entities prioritize high-frequency patrols and enhanced lighting infrastructure within these residential clusters during peak risk intervals to deter criminal activity.

Individual Solutions: The Real-Time Risk Assessment Interface

While physical deterrents, such as high-grade locks and secure storage units, remain fundamental, residents currently lack a dynamic, data-driven mechanism to quantify environmental risk at the point of storage. To address this, a **conceptual** End-User Application is proposed to translate the model's complex historical patterns into real-time, actionable intelligence for the public.

To transition the high-fidelity historical model into a functional public tool, the proposed application logic is designed as follows:

- **Inference Logic for Pre-Incident Scenarios:** To ensure the framework remains applicable in "live" situations, the proposal excludes post-facto variables (such as PRIMARY_OFFENCE, DIVISION) that are unavailable before a theft occurs. This ensures the predictive output relies solely on antecedent predictors available at the moment of use.

- **Automated Feature Ingestion:** The proposed system would reduce user friction by automatically capturing temporal metadata (hour, day, and day of week) via the device's internal clock, maintaining high model accuracy without requiring manual data entry.
- **Proposed User Telemetry:** The interface is designed to focus on a concise set of user-provided inputs, including bicycle specifications (make, model, valuation) cross-referencing these against historical benchmarks to generate an instantaneous risk classification.

BikeGuard: Real-Time Theft Risk Assessment

Purpose

BikeGuard provides cyclists with instantaneous, data-driven risk assessments to help them make informed parking decisions. By analyzing historical theft patterns against specific bike attributes and environmental factors, the app classifies theft risk levels in real-time, allowing users to take proactive security measures.

Core Input Features

To generate a prediction, the user provides five key data points:

- Make & Model: Identifies if specific brands or styles are currently trending targets for theft.
- Colour: Incorporates visual data that may influence a thief's choice.
- Estimated Cost: Evaluates the "reward" factor; higher-value assets often carry different risk profiles.
- Premise Type: Categorizes the parking environment (e.g., apartment, transit station, or street) to assess location-based vulnerability.

Inference Logic & Model Integrity

The app's intelligence is powered by a Random Forest machine learning model. During development, it was discovered that including post-theft data (such as police report dates, primary_offence, or exact coordinates of a crime) created Target Leakage, leading to artificially high accuracy.

To ensure real-world utility, the model has been strictly trained only on precursor data. It ignores any variables that are only known *after* a theft has occurred. This ensures that the model provides a true "pre-crime" risk assessment based solely on the user's current situation.

Automated Feature Ingestion

To maximize accuracy while minimizing user effort, the app automatically injects temporal metadata from the device's internal clock to determine the time of day (Morning, Afternoon, Evening, Night)

User Interface

Input: A streamlined interface for entering bike and premise details.

Output: A risk level of the bike being stolen

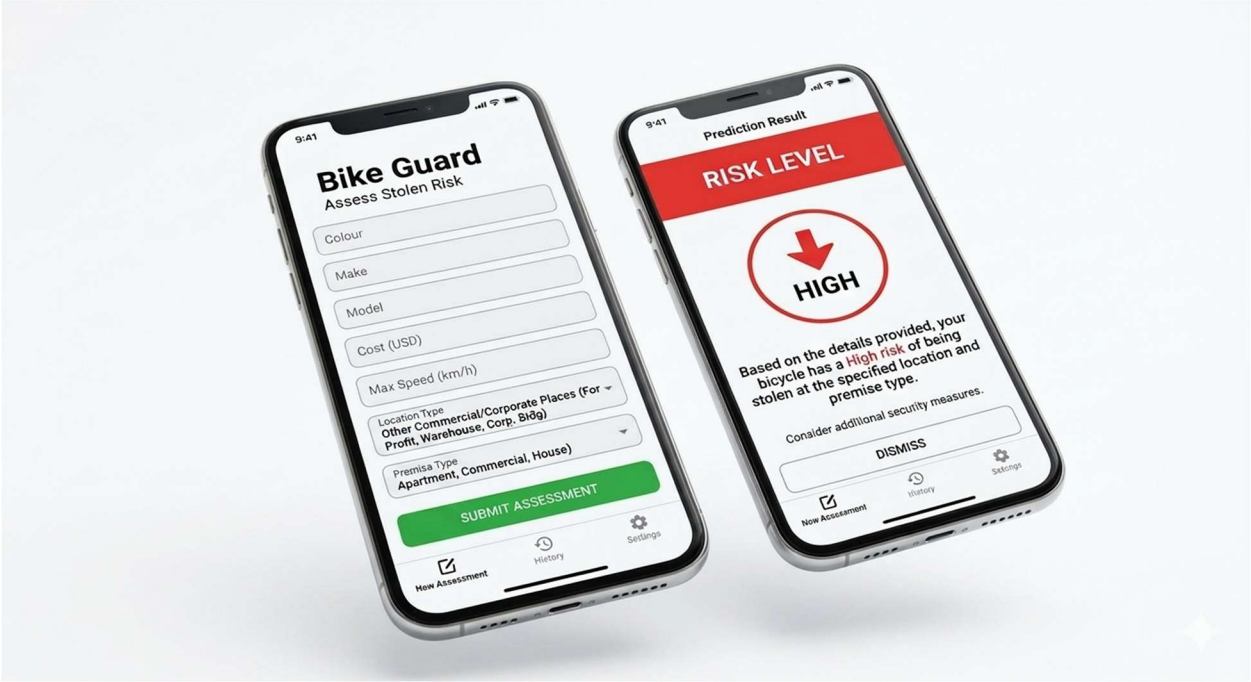

Figure 2- Mock-ups of App

User Benefits

BikeGuard shifts the focus from recovery to prevention. By providing an immediate "Risk Score", users are empowered to change their behavior- such as choosing a different premise type- before a theft occurs.

Future Enhancements

Risk & Mitigation Screen: A visual summary showing the calculated Risk Level (Low, Medium, High) paired with actionable advice (e.g., "Use a D-lock" or "Move to a high-traffic area") to lower that risk.

Historical Trends Report: A feature allowing users to view localized theft statistics based on their specific bike profile.

Hotspot Mapping: Integrating safe-zone recommendations based on historical data.

Technologies Used

The following technologies were employed in this project:

1. MongoDB (NoSQL)
2. SQLite (RDBMS)
3. Apache Spark
4. Spark Structured Streaming
5. Oracle VirtualBox
6. Ubuntu

Key Project Takeaways

1. Data wrangling is critical for extracting valuable insights and enhancing machine learning effectiveness.
2. Model comparison emphasizes the importance of careful feature selection and the risks of data leakage.
3. A comprehensive big data pipeline integrates cold processing, machine learning, and hot streaming.

Conclusion

This project successfully implements a big data pipeline leveraging the tools learned throughout the course and applying them to a production dataset. The analysis clearly shows that the highest volume of thefts occurs at night in apartment complexes along the waterfront. Consequently, the city of Toronto can enhance its policing strategies in these high-risk areas and utilize the “Bike Guard” app to help users assess risk by inputting vital bike details such as colour, make, model, and cost.

References

Sharma, D. (2026).

Week09_W26_Big_Data_System_Design_Lecture_Big_Data_Machine_Learning

[PowerPoint slides].https://learn.ontariotechu.ca/courses/37584/modules